MIL-E-62045B(AT)
14 May 1973
SUPERSEDING
MIL-E-62045A(AT)
8 August 1967, and
ATPD-2045
7 October 1971

MILITARY SPECIFICATION

ENGINE, DIESEL: THREE CYLINDER IN-LINE-TYPE, 90 MINIMUM HORSEPOWER

1. SCOPE

1.1 Scope. This specification covers a three cylinder in-line-type, liquid-cooled, two-stroke-cycle, internal-combustion, and compression ignition engine for use in military vehicles.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

O-A-548	- Antifreeze, Ethylene Glycol, Inhibited.
VV-F-800	- Fuel Oil, Diesel.

Military

MIL-P-514	- Plates; Identification, Instruction and Marking, Blank.
MIL-L-2104	- Lubricating Oil, Internal-Combustion Engine, Tactical Service.
MIL-L-10295	- Lubricating Oil, Internal-Combustion Engine, Sub-Zero.
MIL-B-11188	- Batteries, Storage: Lead-Acid.
MIL-A-11755	- Antifreeze, Arctic-Type.
MIL-E-13856	- Electrical Components for Automotive Vehicles; Water-proofness Tests.

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MIL-L-21260 - Lubricating Oil, Internal-Combustion Engine, Preservative and Break-in.
MIL-A-46153 - Antifreeze Ethylene Glycol, Inhibited, Heavy Duty, Single Package.

STANDARDS
Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U. S. Military Property.
MIL-STD-193 - Painting Procedures, Tactical Vehicles (Tracked and Wheeled).
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.

DRAWINGS
11601645 - Engine (Complete).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U. S. DEPARTMENT OF COMMERCE
Publication No. 235 - Weather Bureau Psychrometric Tables.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington D. C. 20402.)

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ENVIRONMENTAL PROTECTION AGENCY, MOTOR VEHICLES
Federal Register, Volume 35, Number 219.

(Application for copies should be addressed to Environmental Protection Agency, Office of the Administrator, Washington D. C. 20201.)

3. REQUIREMENTS

3.1 Construction.

3.1.1 Assembly. Assembly shall be in accordance with drawing 11601645. All parts requiring identification shall be identified in accordance with MIL-STD-130.

3.1.2 Accessories and equipment. All electrical accessories and equipment, including wiring and electrical connections, shall conform to the applicable requirements of MIL-E-13856. Unless otherwise specified, all accessories and equipment shall be installed on the engine and properly adjusted.

3.1.3 Interchangeability of parts. Component assemblies and parts of the engine shall be so constructed that any parts, except those furnished in matched sets or for which a selective fit is specified, may be installed, replaced, and adjusted without requiring modification.

3.2 Reliability, endurance, and performance requirements.

3.2.1 Reliability. The engine shall be of such quality and workmanship to demonstrate its durability for 20,000 miles of wheeled vehicle operation, without overhaul or major failure (see 6.9) under the following conditions:

(a) Engine operating within the performance requirements specified herein.
(b) Engine operating within the operational requirements specified herein.
(c) With military maintenance as specified in applicable technical manuals, such as changing oil, filters, and necessary adjustments.

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3.2.2 Endurance run. The bare engine (see 6.3) shall be subjected to an endurance run as specified. At the conclusion of the run, the engine shall conform to all the performance requirements of this specification with the exception that the power and torque shall be not less than 95% of initial readings without readjustment of the injection system. In addition, there shall be no evidence of abnormal wear, such as scuffing or galling of cylinder bores, pistons, piston rings, piston pins, bearings, bearing journals, gear teeth, cam surfaces, valves, tappets or other parts, burning of valves or pistons, sticking of valves or piston rings, leakage of gaskets or seals which may result in failure of a component or other malfunction.

3.2.2.1 First production engine. The first production engine, and all engines produced thereafter shall meet the operational and performance requirements of this specification. The first engine produced from production tooling shall be fully representative of engines to be supplied under contract, and shall be furnished to the Government for examination and testing purposes, to, determine conformance to the requirements of this specification. All the engines shall meet the operational and performance requirements specified herein, when using fuels conforming to VV-F-800.

3.2.3 Performance requirements.

3.2.3.1 Speed range. The engine shall operate and meet all performance requirements under all loads and conditions as specified herein through a speed range of 1200 to 2800 rpm. The engine shall maintain an idle speed of between 575 and 625 rpm when using diesel fuel as specified herein.

3.2.3.1.1 Governor. The governor shall limit the maximum engine speed as follows:

Full load operation	2785 to 2835 rpm
No load operation	2940 to 2990 rpm

The engine speed with no load on the engine shall stabilize within 3 seconds after full-rack position is reached. The governor shall have been set prior to shipment.

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3.2.3.2 Brake horsepower. After completion of the production break-in run and the 50 hour run, the bare engine (see 6.3) shall develop not less than 90 observed gross horsepower nor more than 98 observed gross horsepower at 2800 rpm when using fuels specified herein.

3.2.3.2.1 Rating conditions. Engine rating shall be achieved under the following conditions:

Dry air barometer	28.00 to 29.5 inches hg.
Inlet air temperature	70°F to 100°F
Fuel (S.G. 0.853 at 60°F)	70°F to 100°F
Exhaust back pressure	0 to 1 in. hg.
Inlet restriction	0 to -5 in. H_2O

3.2.3.3 Torque. After completion of the production break-in run and the 50 hour run, the bare engine shall develop a minimum of 190 pounds feet observed torque at 1750 rpm.

3.2.3.4 Exhaust smoke density. The maximum exhaust smoke density at full power position and measured within 3 feet of the exhaust manifold outlet shall be not more than position number 6 on a smoke meter (see 6.5).

3.2.3.4.1 Air pollution control. All engines shall comply with the Department of Health, Education, and Welfare Regulations governing control of air pollution from new motor vehicle engines in effect on the date of manufacture.

3.2.3.5 Fuel. The engine shall be capable of full performance when using fuel conforming to Grade DF-2 of VV-F-800.

3.2.3.5.1 Fuel consumption. The fuel consumption of the bare engine at minimum, brake horsepower of 90 and using fuel conforming to Grade DF-2 of VV-F-800, shall be not more than the following:

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Specific fuel consumption at maximum brake horsepower

RPM	Lbs Per PHP-HR
1750	.500
2800	.480

3.2.3.6 Oil. Engine shall perform as specified herein using lubricating oil in accordance with the seasonal requirements of MIL-L-2104 (see 6.8) from minus 10° to plus 115°F., and MIL-L-10295 from minus 65° to 0°F. Oil conforming to type II of MIL-L-21260 may be used during engine buildup and production breakin run specified herein. Referee Grade MB901 of MIL-L-2104 oil shall be used during the 400-hour endurance run (see table II).

3.2.3.6.1 Oil consumption. After 25 hours of operation, the engine shall consume not more than 0.004 lbs. of lubricating oil per observed brake horsepower hour, when operating at full power throughout the speed range of the engine with engine coolant outlet temperature of 130° to 230°F and oil sump temperature of 140° to 250°F.

3.2.3.6.2 Oil pressure. The engine shall operate under all operating conditions specified herein, including low oil level sump condition as indicated by the "add" mark on the oil level dip stick. With the engine operating at 2800 rpm, using OE 30 oil and oil temperature of between plus 180°F. to plus 275°F. the gallery oil pressure shall be not more than 80 psig nor less than 32 psig measured at the pressure sending unit. With the engine operating at specified idle speed, the oil pressure shall be not less than 5 psig.

3.2.3.7 Inlet restriction and exhaust back pressure.

3.2.3.7.1 Inlet restriction. The engine power shall be reduced not more than 2 1/2% with inlet restriction, measured at engine air intake, of 25 in. H_2O gauge, at an engine speed of 2800 rpm, under full load.

3.2.3.7.2 Exhaust back pressure. The engine shall perform as specified with exhaust back pressure, measured at the manifold outlet, of 4.0 in. Hg. gauge, at an engine speed of 2800 rpm under full load and power loss shall be not more than 2 1/2 percent.

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3.2.3.8 Limiting operating temperatures. With the engine operating throughout the speed and load range under any of the conditions specified herein, and an engine coolant outlet temperature of 230°F, the engine oil sump temperature shall not exceed 275°F. During this test, coolant system pressure shall be not more than 31.0 psia.

3.2.3.9 Cooling system. With cap off and a water pump inlet restriction of not more than 3" Hg vacuum, and engine outlet pressure of not more than 10 psig, the engine thermostat in the wide open position, and an engine inlet temperature of 195 ± 5°F, the coolant temperature rise shall not be more than 15°F with the engine operating at 2800 rpm at 90 minimum horsepower. This requirement shall be met with all mixtures of water containing from 0 to 50% antifreeze conforming to MIL-A-11755.

3.2.3.9.1 Coolant. The engine shall perform as specified herein using water and antifreeze compound conforming to O-A-548, arctic-type antifreeze conforming to MIL-A-46153 and artic-type antifreeze conforming to MIL-A-11755, without damage to the engine.

3.2.3.10 Enviromental requirements.

3.2.3.10.1 Starting. The engine shall start within 2 minutes and shall demonstrate the performance characteristics specified herein under any of the following conditions or possible combination of conditions:

a. With integral cold starting aids, and without external aids after being cold soaked without benefit of solar radiation, to an ambient temperature of -25°F. (Cold soak shall be defined as reducing the temperature of the engine and its fuel, coolant and lubricant to within 5°F of the ambient temperature.) Electrical power shall be supplied by two fully charged 6 TN batteries conforming to MIL-B-11188 that were cold soaked with the engine and vehicle electrical system down to within 5°F of the ambient temperature specified.

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b. With the aid of a properly functioning authorized engine coolant heater (arctic kit) under standby operations up to 16 hours at an ambient temperature of -65°F without benefit of solar radiation. (Cold soak shall be defined as reducing the temperature of the engine and its fuel, coolant, lubricating oil and batteries to within 5°F of the ambient air temperature surrounding the engihe.)

c. Without external aids, with cooling fan installed, and a fuel pressure of 2 pounds per square inch gage (psig) at the inlet to the fuel injection supply pump, when operated at the following temperatures with exposure to maximum solar radiation:

Ambient air temperature	115°F.
Intake air temperature (at air cleaner inlet)	130°F.
Fuel temperature (at pump inlet)	130°F.
Coolant outlet temperature	230°F.
Oil sump temperature	At maximum temperature attainable up to 275°F.

d. Humidity conditions. At 29.92 dry barometer inches of mercury and under relative humidity conditions as low as 5 percent at temperature of 115°F. and as high as 100 percent at all temperatures from -25° to + 85°F.

3.2.3.10.2 Elevation conditions. At 8,000 feet elevation, the observed power output of the engine shall be not less than 75 percent of rated power at 2800 rpm's.

3.2.3.11 Grades and slopes. The engine shall start and demonstrate the performance characteristics specified herein when operated throughout the speed range of the engine, for not less than 30 minutes in each direction, with full and low oil level conditions in the sump, as indicated by the "full" and "and" marking on the oil level dip stick, on longitudinal grades up to 60 percent and on lateral slopes up to 40 percent, or any combination thereof. As result of said operation, no evidence of faulty lubrication, leakage, or other malfunction shall be found.

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3.2.3.12 Steam and water jet cleaning. The engine and all its components shall withstand cleaning with high pressure steam, cleaner, and water jet, without deterioration of seals or hose, water leakage past seals or gaskets, and other defects. Paint removal shall not be considered a deficiency. Steam and water jet cleaning is defined as follows:

> The jet is applied perpendicular to the surface being cleaned at a distance of not more than 1 ft. from the surface for steam cleaning, and not more than 5 ft. from the surface for water jet cleaning, and cleaned at a rate of 1 square foot per minute. The jet pressure shall be not less than 100 psi, and not more than 110 psi.

3.3 Production engines. All engines procured under this specification shall meet the operational and performance requirements specified herein.

3.3.1 Production break-in run. Engine, using fuel conforming to Grade DF 2 of VV-F-800, shall be given a break-in run by the contractor to assure the engine as submitted to the Government for acceptance shall meet all requirements specified herein with no further break-in operation.

3.3.1.1 Airbox heater. Prior to engine delivery the airbox heater shall be checked for function. With not less than 12 volts applied to the airbox heater, and using fuels specified herein, it shall ignite, and function as specified on 11601645.

3.3.2 Fifty hour-run. An engine shall be selected from production and operated on fuel conforming to Grade DF-2 of VV-F-800 for 50 hours.

3.4 Radio noise suppression. All engines shall meet the tactical vehicle requirements of MIL-STD-461 except the starter and heater air pump.

3.5 Log sheet. The contractor shall maintain a log sheet of each engine run during all tests specified herein which are performed by the engine manufacture. The form of the log sheet and number of legible copies to be furnished to the Government shall be as approved by the procuring activity.

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3.6 Exterior surface treatment. All exposed exterior surfaces of the engine and its components shall be cleaned, painted, or treated for corrosion resistance as specified on the applicable drawings, or if not so specified, in accordance with applicable provisions of MIL-STD-1193.

3.7 Nameplate. Unless otherwise shown on the applicable drawings, the nameplate, date, and instruction plates shall conform to MIL-P-514

3.8 Leakage. During dynamometer operation, the engine shall show no evidence of leakage greater than a seep as defined in 6.6.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract of purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality control system. Unless otherwise specified by the procuring activity, the contractor shall provide and maintain an effective Inspection and Quality Control System acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality control procedure. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of control required by this specification or other applicable documents referenced therein.

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4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of surveillance of the operations to determine that practices, methods, and procedures of the written inspection plan are being properly applied; and Government product inspection to measure quality of product, offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an effect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Location for tests. All testing referenced in this specification shall be performed at the location specified in Table I.

Table I. Location for test

Title	Requirement Para. No.	Test Para. No.	Place of Manufacture	Government Laboratory
First production inspection and testing	3.2	4.2		X
Endurance test	3.2.2	4.2.1		X
Failure		4.2.1.1		X
Power check	3.2.2	4.2.1.2	X	X
Engine shut down	3.2.2	4.2.1.3		X
Engine warm up period	3.2.2	4.2.1.4		X
Test hours		4.2.1.5		X
Data		4.2.1.6		X
Warning & shut down		4.2.1.7		X
Servicing		4.2.1.8		X
Test temperature		4.2.1.9		X
Performance test	3.2.3.1 thru 3.2.3.9	4.2.2	X	X

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Table I. Location for tests (cont'd)

Title	Requirement Para. No.	Test Para. No.	Place of Manufacture	Government Laboratory
Oil consumption test	3.2.3.6.1	4.2.2.1 & 4.4.3.7	X	X
Cold starting	"a" of 3.2.3.10.1	4.2.2.2	X	X
High temperature starting	"c" of 3.2.3.10.1	4.2.2.3	X	X
Humidity conditions & coolant temperatures	3.2.3.10.1 & 3.2.3.8	4.2.2.4	X	X
Elevation test	3.2.3.10.2	4.2.2.5		X
Grades & slopes test	3.2.3.11	4.2.2.6		X
Steam & water jet cleaning test	3.2.3.12	4.2.2.7		X
Examinations		4.3.1	X	X
Production engine test	3.3 & 3.2.1	4.4	X	
Tear-down inspection		4.4.1.1	X	
Corrections & reassembly		4.4.1.2	X	
Speed range check	3.2.3.1	4.4.2.1	X	
Governor check	3.2.3.1.1	4.4.2 2	X	
Brake horsepower check	3.2.3.2	4.4 2.3	X	
Oil pressure check	3.2.3.6.2	4 4 2 4	X	
Airbox heater check	3.3.1.1	4 4 2.5	X	
Fifty hour control test		4.4 3	X	
Temperatures		4.4.3.3	X	
Torque & brake horsepower check	3.2.3.2 & 3 2.3.3	4 4.3.4	X	
Exhaust smoke density check	3.2.3.4	4.4.3.5	X	
Fuel consumption check	3 2.3.5.1	4.4.3.6	X	
Radio noise suppression test	3.4	4 4.4	X	
Inspection for preparation of delivery		4.5	X	

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4.2 First production inspection and testing. To determine conformance to 3.2, the bare engine shall be subjected to an endurance test and operational tests as defined in 4.2.2, and performance tests defined in 4.2.2 through 4.2.3.7 and examinations specified in 4.3.1. At the conclusion of testing, the engine will be disassembled for tear-down inspection for conformance to applicable sections of this specification. This testing and inspection will be performed by and at a place designated by the government. The contractor shall support the testing by providing any components necessary for completion of the test. Delay in test completion caused by engine breakdown or failure of the contractor to support the engine with parts during testing shall not be the basis for adjustment of contract delivery schedule or contract price.

4.2.1 Endurance test. The endurance test shall consist of a total of 400 hours divided into four periods of 100 hours each. Each 100 hour period shall consist of twenty, 5 hour endurance schedules (see Table II).

Table II. Endurance run five hour schedule

Period	%Rated Speed	%Load	Time, Hrs.
A	Performance	Tests	As Required
1	Idle	0	1/2
2	Max Torque	100	1
3	100	0	1/2
4	75	85	1
5	100	50	1/2
6	100	100	1
7	50	25	1/2
		Total Time, Periods 1-7 - 5 hrs.	

4.2.1.1 Failure. Failure of a first production engine to conform to any of the requirements specified, or any major deficiency of a workmanship material or design nature found as a result of the test shall be cause for rejection, and for refusal to continue acceptance of production engines until objective evidence has been provided by the manufacturer that corrective action has been taken to eliminate the deficiency. Correction of any deficiency must be accomplished by the manufacturer at no cost to the Government on all engines accepted

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under the contract. Any deficiencies found as a result of test is considered prima facie evidence that engines already accepted are similarly deficient unless evidence satisfactory to the contracting officer is furnished by the contractor that no deficiencies exist.

4.2.1.2 Power check tests. To insure engine compliance to performance requirements listed below, the engine shall be subjected to power checks before start of the endurance run and thereafter every 100 hours of endurance time:

Speed Range	(see 3.2.3.1)
Brake Horsepower	(see 3.2.3.2)
Torque	(see 3.2.3.3)
Exhaust Smoke Density	(see 3.2.3.4)
Fuel Consumption	(see 3.2.3.5.1)
Oil Pressure	(see 3.2.3.6.2)

The power checks will be performed with the throttle set in the wide open position (full load) and engine speed of 1600, 2000, 2400, and 2800 RPM in both ascending and descending order. At each engine setting, the engine should be run for a sufficient time (approximately 20 minutes) for stabilization.

4.2.1.3 Engine shut down. The engine test may be shut down after eompletion of any period in the five hour schedule. Planned shut down following test periods 2, 4, and 5 will include a cool down running of period No. 7. This cool down period will not be credited as endurance test time. This cool down period is not required after periods 1, 3, and 7.

4.2.1.4 Engine warm-up period. A one half hour warm-up period is required prior to continuation of endurance run following any shut down. During this one half hour period, the loads and speeds shall be progressively increased to the values required in period No. 5. Temperature stabilization in period No. 5 is required. This warm-up period will not be credited as endurance time.

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4.2.1.5 Test hours. Any period of the schedule terminated due to equipment malfunction or minor test speciment failure must be repeated when time elapsed is less than half. The voided part of the period will not be accrued as endurance time.

4.2.1.6 Data. The following data will be recorded during the last five minutes of each schedule periods and power check tests:

(a) Engine speed (RPM).
(b) Observed brake horsepower (OBHP) and/or torque (lb-ft).
(c) Fuel flow (lbs/hour).
(d) Test cell ambient temperature (^{o}F) and barometric pressure (in. Hg).
(e) Engine oil gallery and sump temperatures (^{o}F) and pressure (PSIG-Gallery only).
(f) Fuel secondary filter out and spill back temperatures (^{o}F) and pressures (PSIG).
(g) Air cleaner inlet temperature (^{o}F).
(h) Air inlet pressure (in. Hg).
(i) Air cleaner restriction (in. water).
(j) Coolant engine inlet and outlet temperatures (^{o}F) and pressures (PSIG).
(k) Exhaust port temperatures (^{o}F).
(l) Exhaust manifold pressure (in. Hg).
(m) Crankcase pressure (in. water).
(n) Fuel, transfer pump, inlet pressure (PSIG).
(o) Oil consumption (lb/hr) will be calculated on basis of oil added in lbs accumulating to the engine as necessary during the endurance run.
(p) Exhaust smoke density (power check only).

4.2.1.7 Warning and shut downs. To guard against overstress caused by equipment malfunction or human error, the following automatic warning shutdown provisions will be incorporated in each test cell:

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a. Temperatures °F, maximum	WARNING	SHUT DOWN
(1) Coolant, engine out	225	235
(2) Oil, engine sump	270	280

b. Pressure PSIG, minimum	WARNING	SHUT DOWN
(1) Oil engine gallery 600-1400 RPM	8	5
(2) Oil engine gallery 1400-2200	18	10

c. Engine speed

4.2.1.8 Servicing. Prior to the power check test, the lubricating oil shall be changed, all filters serviced, valve clearance and injector rack linkage checked and adjusted to applicable requirements.

4.2.1.9 Test temperatures. The endurance test, including power checks, will be performed at an ambient cell temperature of 85 ± 5°F, except for the last 100 hours which will be performed at an ambient cell temperature of 125 ± 5°F.

4.2.2 Performance tests. To determine conformance to 3.2.3.1 through 3.2.3.9, with the engine operating, as specified, it shall be checked for meeting the performance requirements during the endurance run.

4.2.2.1 Oil consumption check. To determine conformance to 3.2.3.6.1, during the endurance run, the engine shall be checked for oil consumption.

4.2.2.2 Cold starting. To determine conformance to item "A" of 3.2.3.10.1, the bare engine, with integral winterization aids, shall be cold started at -25°F. on fuel conforming to the seasonal requirements of VV-F-800. Prior to the cold start, the engine and batteries shall be cold-soaked down to within a ± 5°F ambient temperature of -25°F. After each start, the engine must operate for a sufficient time to stabilize engine operation.

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4.2.2.3 High temperature starting. To determine conformance to item "c" of 3.2.3.10.1, the bare engine, without aids, shall be placed in a high temperature chamber maintained at a temperature of 115°F., and operated at full load until all temperatures are stabilized within the limits specified. The engine shall then be shut down, hot soaked, and restarted when the fuel temperature at the inlet to the fuel transfer pump reaches its maximum value. Fuel shall be supplied to the engine mounted fuel supply pump at a pressure of 5 psi. The fuel pressure shall be shut off during the hot soak period.

4.2.2.4 Humidity conditions and coolant temperatures. To determine conformance to item "D" of 3.2.3.10.1 and 3.2.3.8, the engine shall be operated under actual or simulated humidity conditions and checked for limited operating temperatures.

4.2.2.5 Elevation test. To determine conformance to 3.2.3.10.2, the engine shall be placed in a simulated enviromental condition of 8,000 feet elevation, and its general performance characteristics observed.

Elevation feet	*Pressure inches of mercury (Absolute)	Temperature °F (Air Cleaner Inlet)

4.2.2.6 Grades and slope test. To determine conformance to 3.2.3.11, the initial production engine shall be subjected to this test after completion of the erdurance test.

4.2.2.7 Steam and water jet cleaning test To determine conformance to 3.2.3.12 after the completion of the 400-hour endurance test, and just prior to the tear-down inspection, the engine shall be subjected to normal steam cleaning with cleaner as specified herein. Immediately upon completion of the steam cleaning operation, the engine shall be subjected to normal water jet cleaning as specified. Prior to cleaning, the air intake opening, flywheel, exhaust, and all other openings that might be affected shall be covered. The engine shall start and operate at idle speed without misfiring within two minutes immediately after the completion of both the steam and water

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jet operations. During disassembly of the engine, special notice should be made to water entering the engine through joints, seals, gaskets. Evidence of malfunction or of water seepage past seals, gaskets, and joints, shall constitute a failure of the test.

4.3 Inspection provisions.

4.3.1 Examinations. Visual and primary functional examination shall consist of examination of the engines for conformance to the applicable drawings and this specification. Examination shall be performed against the classification of defects and with the acceptable quality levels shown in Table III. The following constitutes a part of the classification of defects (see 6.7):

(a) Any drip from the inside to the outside constitutes a major defect when the engine is not in operation and the components are at ambient temperature.
(b) Any seep that occurs from the inside to the outside of a static fit (metal to metal, or gasket combination) shall constitute a major defect.

Table III. Classification of defects

Characteristic		Major AQL Defects = 6.5 percent	Method of Inspection
101.	Valve tappet clearance each valve	Exhaust valves improperly adjusted	Gage
102.	Oil sump, fuel and oil filters	Dirt, chips and foreign particles, due to production	Visual and functional
103.	Fuel, oil, and coolant leaks	Leakage	Visual
104.	Fuel lines	Damaged	Visual and functional
105.	Governor	Malfunction	Visual and functional

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Table III. Classification of defects (cont'd)

	Characteristic	Major AQL Defects = 6.5 percent	Method of Inspection
106.	Torque on cylinder head bolts, exhaust manifold flange bolts, vibration damper	Improper torque	Torque wrench
107.	Crankshaft	Excessive or restrictive play	Gage and functional
108.	Engine	Malfunction	Functional
109.	Fuel system components	Malfunction, damage or leaks, etc.	Visual and functional
110.	Fuel, injection system, timing and components	Malfunction, improper adjustment	Gage
111.	Coolant system	Malfunction nonconformance	Functional
112.	Airbox pressure	Less than limits	Gage
113.	Crankcase pressure	Exceeds limits	Gage

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Table III. Classification of defects

	Characteristic	Minor AQL Defects = 10.0 percent	Method of Inspection
201.	Linkage	Improperly adjusted	Visual
202.	Minor assemblies	Omitted	Visual
203.	Engine	Improper adjustment or installation of components	Visual and functional
204.	Fuel lines, hose, verts, shut-off valve	Improper assembly or installation	Visual
205.	Painting	Improper application	Visual
206.	Workmanship	Not following good practice, improper installation or adjustment of components	Visual and functional

4.3.1.1 Unclassified defects. All component defects having no bearing on function, safety, interchangeability or life, but which are considered departures from good workmanship, will be noted in writing. Workmanship deficiencies failing within this category and recurring in 5 consecutive lots, or in 10 lots or more within a thirty-day period, will be added to the minor classification with no increase to AQL's.

4.3.1.2 Recurring major deficiencies. A major deficiency (see 6.7) is recurring when the same defect occure more than once in the same sample or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or approved equivalent) reflect such a condition. Recurring major deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies The deficiencies shall be corrected when found.

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4.3.1.3 Recurring minor deficiencies. A minor deficiency (see 6.7) is recurring if it occurs more than twice in the same sample or when the defect occurs in four successive samples. Recurring minor deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. Correction of all deficiencies shall be accomplished prior to acceptance.

4.4 Production engine test. To determine conformance to 3.3 and 3.2.1, each engine shall be operated in accordance with 3.3.1. Any unsatisfactory operation or conditions which require correction shall be corrected by the contractor before the engine is presented for acceptance.

4.4.1.1 Tear-down inspection. At the conclusion of the break-in run, and before acceptance by the Government, the contractor, in the presence of the Government inspector, shall disassemble and inspect only the first 10 engines, and every second engine of the next 10 before sampling may be allowed. Disassembly shall be accomplished to the extent required to perform the following inspection:

(a) Oil contamination.
(b) Dirt, chips or foreign matter in the engine block, oil pan, filters, and accessory drive case.
(c) Main bearing bolt torque.
(d) Connecting rod bolt torque.
(e) Cylinder bore scuffing, scoring, galling.
(f) Piston scuffing and burning.
(g) Rod and main bearing condition.

4.4.1.2 Corrections and reassembly. Should there be evidence of any condition of engine or parts requiring correction, such corrections shall be made by the contractor prior to reassembly of the engine and re-submission to the break-in run. In addition, the contractor shall submit evidence to the Government, that corrective action has been taken in production to eliminate any deficiency disclosed. Failure of the contractor to submit such evidence shall be cause for refusal by the Government to continue acceptance of subsequent engines. Any engine requiring replacement parts shall be reassembled and subjected to the break-in run and acceptance power check test. Any engine not requiring replacement parts shall be reassembled and subjected to the acceptance power check test only

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4.4.2 Acceptance tests. After break-in run, each engine shall be observed for conformance to the operational and performance requirements.

4.4.2.1 Speed range check. To determine conformance to 3.2.3.1, with the engine operating under all load conditions, it shall be observed for meeting speed ranges.

4.4.2.2 Governor check. To determine conformance to 3.2.3.1.1, with the engine operating under full load and no load, it shall be observed for meeting speed requirements.

4.4.2.3 Break horsepower check. To determine conformance to 3.2.3.2, with the engine operating as specified, it shall be observed for meeting the brake horsepower requirements.

4.4.2.4 Oil pressure check. To determine conformance to 3.2.3.6.2, with the engine operating, using specified oil and at indicated temperature, the oil pressure shall be not more than that specified.

4.4.2.5 Airbox heater check. To determine conformance to 3.3.1.1, during acceptance, endurance, and 50-hour runs, the airbox heater shall have the specified voltage applied and checked for ignition requirements.

4.4.3 Fifty-hour quality control test. A 50-hour test (Table IV) shall be conducted by the contractor as specified herein for the purpose of determining the continued quality of the engine and to establish a measure of reliability. Following the 50-hour control test, the engine shall be subjected to a power check test to insure compliance to performance requirements listed below:

Speed range	(see 3.2.3.1)
Brake horsepower	(see 3.2.3.2)
Torque	(see 3.2.3.3)
Exhaust smoke density	(see 3.2.3.4)
Fuel consumption	(see 3.2.3.5.1)
Oil pressure	(see 3.2.3.6.2)

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The power checks will be performed with the throttle set in the wide open (full rack) position at engine speeds of 1600, 2000, 2400, and 2800 RPM, in both ascending and descending order. At each interval, the engine shall be run for a sufficient time (approximately 20 minutes) for stabilization. After conclusion of this test, the engine shall be disassembled and inspected in accordance with 4.4.1.1 tear down inspection.

4.4.3.1 Selection of test sample. Engines shall be selected by the government at the rate of one per month when production is 100 units per month or less, and at the rate of two per month when production is greater than 100 units per month. Engines shall be identified as to production period.

4.4.3.2 Control test failure. Failure of a control test sample to pass any specified examination or test shall be cause for rejection and for the Government to refuse acceptance of subsequent lots until it has been proven to the satisfaction of the Government that the faults revealed by the test have been corrected.

4.4.3.3 Temperatures. Inlet air temperatures shall be not less than 70°F. nor more than 100°F. The lubricating oil temperature shall not exceed 275°F. The coolant (engine inlet) temperature shall be maintained at 180°F ± 5°F.

Table IV. 50 hour control test

Period	Time (hs)	Speed (rpm)	Brake horsepower	Fuel
1	(run in schedule)	(see 3.5)		
2	5	1600	30	Fuel conforming to
3	5	2000	58	VV-F-800 shall be
4	5	2800	73	used during the en-
5	5	2400	84	tire run
6	30	2800	90	
7	Power check test at minimum of 90 hp (see 4.2.1)			

NOTE: 1. The engine coolant inlet temperature shall be maintained at 180°F ± 5°F.

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2. The barometric pressure shall be recorded during the power check test.
3. Data required (see 4.2.1.6).
4. Use Grade DF-2 of VV-F-800 for power check.
5. Power check test to be run at a full rack position.

4.4.3.4 Torque and brake horsepower check. To determine conformance to 3.2.3.2 and 3.2.3.3, with engine operating as specified, it shall be observed for meeting the brake horsepower and torque requirements.

4.4.3.5 Exhaust smoke density check. To determine conformance to 3.2.3.4, with the engine running at full power, the exhaust shall be checked for smoke density.

4.4.3.6 Fuel consumption check. To determine conformance to 3.2.3.5.1, with the engine operating at 2800 rpm, engine shall be checked for meeting fuel consumption and brake horsepower when using specified fuel.

4.4 3.7 Oil consumption check. To determine conformance to 3.2.3.6.1 during the 50-hour control run, the engine shall be operated for a minimum of 5 hours at speed indicated in period 6 in Table IV and then checked for oil consumption.

4.4 3.8 Corrections. Upon completion of the test specified herein, the engine shall be partially disassembled as in 4.4.1.1. Evidence of component or parts damage indicated by this inspection shall require such further disassembly of the engine necessary to determine the cause of failure and correction.

4.4.3.9 Disposition of engine. After correction of all defects disclosed by the test, engine shall be reassembled, tested as specified in 4.4.2, inspected, and, if found acceptable, shipped as a regular production engine.

4.4 4 Radio noise suppression test. To determine conformance to 3.4, the engine as installed in a vehicle shall be tested at a Government proving ground.

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4.5 Inspection for preparation of delivery. Inspection of preservation, packaging, packing, and marking shall be in accordance with the applicable procedures outlined in section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing, and marking, shall be in accordance with the applicable packaging Data Sheet as furnished at the time of invitation for bids for the level of preservation and packaging, and level of packing specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. The engines covered by this specification are intended for production use, as spares, or as replacements for military tactical and transport vehicles.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of method and applicable level of preservation and packaging, and level of packing of the referenced specification (see 5.1).
(c) Order of precedence, where the requirements of this specification and those of any applicable specification differ, the requirements of this specification shall govern.

6.3 Bare engine. The term "bare engine" is defined as an engine specified on Drawing 11601645.

6.4 Gross brake horsepower. Gross brake horsepower shall be defined as the horsepower output of the bare engine at the flywheel.

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6.5 Exhaust smoke density. Exhaust smoke density shall be determined with a Robert Bosch Model EFAW68 Smoke Meter or equal, and Model EFAW65 Sampling Pump or equal (as approved by Government). At full rack setting (throttle) the engine should develop 90 minimum horsepower at 2800 rpm.

6.6 The following definitions shall be used for the classification of defects for leaks:

(a) Weep: - Any non-recurring evidence of fluid beyond the seal.
(b) Seep: - Any recurring evidence of fluid beyond the seal that does not result in formation of a droplet.
(c) Droplet: - Any recurring evidence of fluid beyond the seal that results in formation of a non-falling droplet.
(d) Drip: - Any recurring evidence of fluid beyond the seal where droplet forms and falls.

6.7 Definitions of recurring major and minor defects. The Government Inspector will verify that a thorough inspection of each vehicle is performed by the contractor not only for the listed characteristics, but also for any other departures from good workmanship. The Government Inspector will assure that all deficiencies encountered during the inspection are enumerated on the Deficiency Sheet for the vehicle. The defects noted on Deficiency Sheet shall contain sufficient description to enable the Government Inspector and the contractor's representative to classify the deficiency in accordance with the classification of defects of the vehicle specification and definitions contained in MIL-STD-105. Corrective action shall be taken for recurring deficiencies.

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6.8 Referee grade oil. The designation for grade of referee oil is as follows:

Grade	Government designation	Manufacturer designation
10	MB901	CHI-139

6.9 Major failures. Major failures are defined as follows.

(a) Main bearing and connecting rod bolt broken, stripped threads, scored, and damage that may cause subsequent failure.
(b) Broken, scuffed and burned pistons.
(c) Cylinder bore scuffed, scored, and galling.

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